Then the diagonal OC is the resultant ($\vec{R}$) passing through the common tail O.

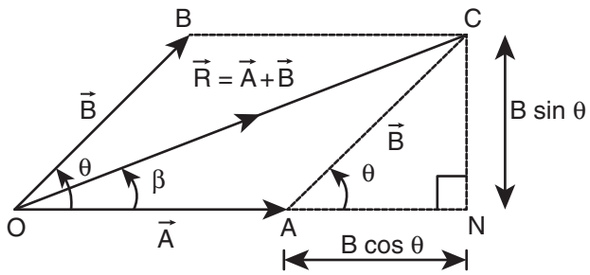


Figure 2.19 Magnitude of resultant vector by parallelogram method

We next find the magnitude and direction of this resultant vector.

(i) Magnitude:

First extend OA to the point N, so that we get ON. Then CN is drawn perpendicular to this ON, from C. Then ONC is a right angled triangle.

We can write $R^2 = ON^2 + CN^2$

$$R^2 = (OA + AN)^2 + CN^2$$

$$\Rightarrow R^2 = A^2 + B^2 + 2AB\cos\theta$$

$$\therefore R = |\vec{R}| = |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

Special cases:

When $\theta = 0°$, then R = A+B

When $\theta = 180°$, then R = A-B

When $\theta = 90°$, then $R = \sqrt{A^2 + B^2}$

(ii) Direction

Let β be the angle between the vectors $\vec{A}$ and $\vec{R}$. Then

$$\tan\beta = \frac{CN}{ON} = \frac{B\sin\theta}{A + B\cos\theta}$$

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